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MODULAR PLATFORM SYSTEM FOR VERTICAL ORE STOPE MINING

Abstract

A modular platform system for vertical ore stope mining, comprising: a top platform having an access opening formed therein for accessing a raise opening of a raise above a stope of a mine; an equipment deck adapted to receive equipment; and, at least one winch having at least one respective cable coupled thereto, the at least one winch mounted on the top platform, the at least one cable extendable through the access opening, the at least one cable having at least one respective cable end coupled to the equipment deck; wherein the at least one winch is operable to move the equipment deck and the equipment between a raised position proximate a top of the raise at the top platform and a lowered position proximate a bottom of the raise at a top of or within the stope.

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Background/Summary

[0001] This application claims priority from and the benefit of the filing date of U.S. Provisional Patent Application No. 63/336, 495, filed Apr. 29, 2022, and the entire content of such application is incorporated herein by reference.

FIELD OF THE APPLICATION

[0002] This application relates to the field of platforms or lifts for mining, and more specifically, to a modular platform system for vertical ore stope mining, underground mining, stope mining, and the like.

BACKGROUND

[0003] As the mining industry is constantly evolving, so are the safety standards required for miners and operators working in mining operations. For example, vertical ore stope mining is often performed using aged or obsolete equipment which places miners and operators at risk as they are in close proximity to blasting equipment, unstable geology, and often work deep below the surface in suboptimal conditions.

[0004] Underground ore mining generally involves the use of a vertical shaft or raise, which accesses horizontal drifts surrounding ore bodies, and a vertical stope in which rock fragmentation via drilling and blasting operations are conducted. Once fragmented, the newly blasted ore is collected and transported through an ore pass and/or back up the vertical shaft or raise to the drift and then on to the surface. Miners, operators, and equipment are typically transported to the drilling and blasting site via a platform, lift, or elevator system. One limitation of effective ore extraction in underground mines is the result of increasing mine depths, which results in increased geotechnical risks, longer travel times through the mine, greater ventilation requirements, and potentially dangerous working conditions for miners and operators.

[0005] In addition, there are several disadvantages of current underground ore mining methods. First, for the correct placement of a vertical stope and blasting equipment, miners and operators must work in close proximity to the equipment, deep underground, which may pose serious safety risks. Second, the efficiency and productivity of ore extraction in underground mines may be decreased due to human error and/or human limitations, more specifically, the time it takes to descend safely to the blasting site in the underground mine, set-up drill and blast equipment, and then to ascend again to a safe location during blasting operations. Third, the requirements for ventilation systems for underground mines are substantially increased and typically require automated systems to meet the needs of multiple sublevels, multiple openings in stopes, including sublevel openings, adequate contaminant removal, and increased system monitoring. Fourth, the infrastructure setup to ensure miner and operator safety with current methods is complex and time-consuming. Fifth, the drill and blast accuracy of current methods is problematic due to human error and/or limitations, and geotechnical challenges. Sixth, safely and quickly accessing a drilling and blasting site by miners and operators with the required equipment in the stope of an underground mine is time consuming and potentially dangerous using existing platform, lift, and elevator systems.

[0006] A need therefore exists for an improved platform system for mining operations. Accordingly, a solution that addresses, at least in part, the above and other shortcomings is desired.

SUMMARY OF THE APPLICATION

[0007] According to one aspect of the application, there is provided a modular platform system for vertical ore stope mining, comprising: a top platform having an access opening formed therein for accessing a raise opening of a raise above a stope of a mine; an equipment deck adapted to receive equipment (e.g., mobile equipment and/or stationary equipment); and, at least one winch having at least one respective cable coupled thereto, the at least one winch mounted on the top platform, the at least one cable extendable through the access opening, the at least one cable having at least one respective cable end coupled to the equipment deck; wherein the at least one winch is operable to move the equipment deck and the equipment between a raised position proximate a top of the raise at the top platform and a lowered position proximate a bottom of the raise at a top of or within the stope.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] Features and advantages of the embodiments of the present application will become apparent from the following detailed description, taken in combination with the appended drawings, in which:

[0009] FIG. **1** is a front perspective view illustrating a modular platform system positioned over a raise in an underground mine in accordance with an embodiment of the application;

[0010] FIG. **2** is a front perspective detail view illustrating the modular platform system of FIG. **1** in accordance with an embodiment of the application;

[0011] FIG. **3** is a front perspective view illustrating the top platform of the modular platform system of FIG. **1** in accordance with an embodiment of the application;

[0012] FIG. **4** is a top view thereof;

[0013] FIG. **5** is a front view thereof, the rear view being substantially a mirror image thereof;

[0014] FIG. **6** is a left side view thereof, the right side view being substantially a mirror image thereof;

[0015] FIG. **7** is a front perspective view illustrating the equipment deck of the modular platform system of FIG. **1** in accordance with an embodiment of the application;

[0016] FIG. **8** is a top view thereof;

[0017] FIG. **9** is a front view thereof, the rear, left side, and right side views being substantially the same;

[0018] FIG. **10** is a front perspective view illustrating the modular platform system of FIG. **1** with the equipment deck thereof shown in a raised position in accordance with an embodiment of the application;

[0019] FIG. **11** is a front perspective view illustrating the modular platform system of FIG. **1** with the equipment deck thereof shown in a lowered position in accordance with an embodiment of the application;

[0020] FIG. **12** is a front perspective view illustrating an equipment deck for the modular platform system of FIG. **1** having a removable cassette deck and with the removable cassette deck thereof shown in an inserted position in accordance with an embodiment of the application;

[0021] FIG. **13** is a bottom perspective view thereof;

[0022] FIG. **14** is a front perspective view illustrating the equipment deck of FIG. **12** with the removable cassette deck thereof shown in a removed position in accordance with an embodiment of the application;

[0023] FIG. **15** is a bottom perspective view thereof;

[0024] FIG. **16** is a front view thereof;

[0025] FIG. **17** is a rear view thereof;

[0026] FIG. **18** is a top view thereof;
[0027] FIG. **19** is a bottom view thereof;
[0028] FIG. **20** is a right side view thereof, the left side view being substantially a mirror image thereof;
[0029] FIG. **21** is a front perspective view illustrating the top deck of the equipment deck of FIG. **12** with the removable cassette deck thereof removed and with the top deck shown in an unfolded position in accordance with an embodiment of the application;
[0030] FIG. **22** is a front perspective view illustrating the top deck of the equipment deck of FIG. **21** with the removable cassette deck thereof removed and with the top deck shown in a folded position in accordance with an embodiment of the application;
[0031] FIG. **23** is a right side view thereof, the left side view being substantially a mirror image thereof;
[0032] FIG. **24** is a front perspective view illustrating the equipment deck of FIG. **12** with the removable cassette deck thereof shown in a removed position and with equipment mounted on the removable cassette deck in accordance with an embodiment of the application; and,
[0033] FIG. **25** is a front perspective view illustrating the equipment deck of FIG. **12** with the removable cassette deck thereof shown in an inserted position and with equipment mounted on the removable cassette deck in accordance with an embodiment of the application.
[0034] It will be noted that throughout the appended drawings, like features are identified by like reference numerals.

DETAILED DESCRIPTION OF THE EMBODIMENTS

[0035] In the following description, details are set forth to provide an understanding of the application. In some instances, certain structures, techniques and methods have not been described or shown in detail in order not to obscure the application.
[0036] The present application provides an elevator-like mobile, modular, automated vertical ore stoping system (“VOSS”) (referred to in the following as a modular platform system **100**). As further described below, the modular platform system **100** includes the following: a mobile, modular frame unit (or top platform **200**) having forklift pockets and easy access lifting lugs; a modular base unit (or equipment deck **500**, **1500**) having remote capability, software integration, and drive on/drive off capability for drill rigs and explosives (e.g., equipment **8000**); modular power units with plug and play cabling; mobile electrical components (e.g., IP 67 rated); LTE (i.e., Long-Term Evolution) wireless or cellular communications to the equipment deck **500**, **1500**; an optional raisebore concrete pad with rails; optional installation of a cap/collar/center line; optional manhole access on the equipment deck **500**, **1500** for 3.sup.rd party stop monitoring; and, an optional telescoping equipment deck **500**, **1500** to provide double deck drilling capability. The modular platform system **100** may have a payload of 20,000 kg and a lift speed of 5-10 m/s according to one embodiment.
[0037] FIG. **1** is a front perspective view illustrating a modular platform system **100** positioned over a raise (e.g., a vertical raise) **5100** in an underground mine **6000** in accordance with an embodiment of the application. And, FIG. **2** is a front perspective detail view illustrating the modular platform system **100** of FIG. **1** in accordance with an embodiment of the application.
[0038] According to one embodiment of the application, there is provided a modular platform system (or system) **100** for mining operations (e.g., underground mining, stope mining, etc.). The modular platform system **100** includes: a top platform **200**; an equipment deck **500** (or **1500** as described further below and as shown in FIG. **12**); one or more cables **400** (e.g., **410**, **420**, **430**, **440**) coupling the top platform **200** to the equipment deck **500**, **1500**; and, a control system **300** for controlling and monitoring the modular platform system **100** and equipment **8000**. The equipment **8000** may include mobile equipment **8000** (e.g., with wheels, with a tracked undercarriage, etc.) and/or stationary equipment **8000** (e.g., without wheels, without a tracked undercarriage, etc.). As further described below, the modular platform system **100** is modular and mobile, that is, the

components **200**, **300**, **400**, **500**, **1500** of the modular platform system **100** may be easily relocated to different locations (e.g., above different raises **5100**) in a mine (e.g., an underground mine **6000**). The modular platform system **100** has a longitudinal axis **110**, a lateral axis **120**, and a vertical axis **130** that may be aligned with a vertical axis **5110** of a raise **5100** above a stope **5000** of the mine **6000**. The top platform **200**, equipment deck **500** (or **1500**), and cables **400** are typically made from metal such as steel.

[0039] FIG. **3** is a front perspective view illustrating the top platform **200** of the modular platform system **100** of FIG. **1** in accordance with an embodiment of the application. FIG. **4** is a top view thereof. FIG. **5** is a front view thereof, the rear view being substantially a mirror image thereof. And, FIG. **6** is a left side view thereof, the right side view being substantially a mirror image thereof.

[0040] According to one embodiment, the top platform **200** has a front end **201**, a rear end **202**, a right side **203**, a left side **204**, an upper or top surface **205**, and a lower or bottom surface **206**. The top platform **200** includes a front platform or deck **210** for use by an operator **7000**, a rear platform or deck **220** (also for optional use by the operator **7000**), a right or first side platform or deck **230**, and a left or second side platform or deck **240**. The front deck **210** has an outer side wall (or side) **211**, an inner side wall (or side) **212**, a first or right side wall or end wall (or end) **213**, and a second or left side wall or end wall (or end) **214**. Similarly, the rear deck **220** has an outer side wall (or side) **221**, an inner side wall (or side) **222**, a first or right side wall or end wall (or end) **223**, and a second or left side wall or end wall (or end) **224**. The right side deck **230** has an outer side wall (or side) **231**, an inner side wall (or side) **232**, a first or right side wall or end wall (or end) **233**, and a second or left side wall or end wall (or end) **234**. Similarly, the left side deck **240** has an outer side wall (or side) **241**, an inner side wall (or side) **242**, a first or right side wall or end wall (or end) **243**, and a second or left side wall or end wall (or end) **244**.

[0041] The right end **213** of the front deck **210** is removably attached (e.g., coupled, bolted, fastened, pinned, etc.) to the left side **234** of the right side deck **230** and the right side **233** of the right side deck **230** is removably attached to the left end **224** of the rear deck **220**. Similarly, the left end **214** of the front deck **210** is removably attached to the right side **243** of the left side deck **240** and the left side **244** of the left side deck **240** is removably attached to the right end **223** of the rear deck **220**. As such, when the front, rear, right side, and left side decks **210**, **220**, **230**, **240** are attached, a raise access opening (or central opening or access opening or opening) **250** is formed in the top platform **200** (i.e., defined by the respective inner sides **212**, **222**, **232**, **242** of the front, rear, right side, and left side decks **210**, **220**, **230**, **240**).

[0042] According to one embodiment, the inner sides **212**, **222**, **232**, **242** of the front, rear, right side, and left side decks **210**, **220**, **230**, **240** (and/or the respective upper surfaces **205** thereof adjacent to the inner sides **212**, **222**, **232**, **242**) are curved or arced outwardly and hence the access opening **250** formed in the top platform **200** is circular (or approximately circular) in shape. Of course, the access opening **250** may be of any shape (e.g., square, oval, rectangular, etc.).

[0043] Referring again to FIGS. **1** and **2**, the top platform **200** is sized to fit over and cover the opening (or raise opening) **5120** at the top **5101** of a raise (or raisebore or bore) **5100** formed in the floor **6300** of a drift **6100** or chamber **6200** above a stope **5000** in an underground mine **6000**. Similarly, the access opening **250** formed in the top platform **200** is sized to match (or approximately match) the diameter **5103** of the raise **5100** (e.g., 4000 mm).

[0044] The top platform **200** is also sized to be moved along a drift **6100** in an underground mine **6000** and assembled in a chamber **6200** over the raise **5100** above a stope **5000**. When assembled, as shown in FIGS. **4** to **6**, the top platform **200** may measure approximately 9092 mm wide by 6568 mm deep by 2687 mm high according to one embodiment.

[0045] According to one embodiment, first and second synchronized winches **261**, **262** may be spacedly mounted on the upper surface **205** of the front deck **210** proximate the right and left ends **213**, **214** thereof, respectively. Similarly, third and fourth synchronized winches **263**, **264** may be spacedly mounted on the upper surface **205** of the rear deck **220** proximate the right and left ends **223**, **224** thereof, respectively. The first, second, third, and fourth synchronized winches **261**, **262**, **263**, **264** are equipped with first, second, third, and fourth cables **410**, **420**, **430**, **440**, respectively, as further described below.

[0046] The synchronized winches **260** (e.g., **261**, **262**, **263**, **264**) may be electrically or hydraulically powered and may have the following ratings according to one embodiment: a capacity of 4,490 Kg; a rope (cable) diameter of 18 m; a speed of 14 m/min; and, a drum storage of 200 m.

[0047] According to one embodiment, first and second ramps **235**, **245** are removably attached to respective outer sides **231**, **241** of the right and left side decks **230**, **240**. The first and second ramps **235**, **245** provide improved access to the equipment deck **500** (or **1500**) for mobile (or stationary) equipment **8000** from the drift **6100**. The ramps **235**, **245** may have a width of 2850 mm according to one embodiment.

[0048] According to one embodiment, the respective upper surfaces **205** of the front and rear decks **210**, **220** have mounted thereon respective safety railings **271**, **272**, safety gates **273**, **274**, **275**, **276**, and canopies **277**, **278** for the protection of operators **7000** working on or near the modular platform system **100**.

[0049] According to one embodiment, the outer sides **211**, **221** of the front and rear decks **210**, **220** may have one or more (e. g., four (4)) leveling jacks **280** spacedly mounted thereto for leveling the system **100** with respect to the floor **6300** of the drift **6100** or chamber **6200** of the mine **6000** where the modular platform system **100** is deployed. In addition, the outer sides **211**, **221** of the front and rear decks **210**, **220** may have one or more (e.g., six (6)) anchoring pads **290** and anchors **291** spacedly mounted thereto for anchoring the modular platform system **100** to the floor of the floor **6300** of the drift **6100** or chamber **6200**.

[0050] According to one embodiment, the control system **300** is mounted on the upper surface **205** of the front deck **210** and is used for local control and monitoring of the modular platform system **100** and its components (e.g., synchronized winches **260**, power supplies **580**, etc.). According to another embodiment, the control system **300** may be remotely located and may be communicatively coupled to the modular platform system **100** and its components over a wired or wireless communications network (e.g., cellular, WIFI, LAN, WLAN, etc.) **351** for remote control and monitoring of the modular platform system **100** and its components.

[0051] The control system **300**, whether located locally or remotely, may be used for the local or remote control of the equipment **8000** that is transported by the modular platform system **100** to and from the stope **5000** via the raise **5100**. The control system **300** may make use of CCTV cameras or the like mounted on the equipment platform **500** (or **1500**) when controlling equipment **8000**. The control system **300** may include one or more local or remote user interfaces, input devices, and output devices (e.g., keyboards, control panels, displays, touchscreens, graphical user interfaces, joy sticks, mice, printers, buttons, etc.) **310** for receiving commands from an operator **7000** and for presenting information (e.g., status indications, measured values, alarms, etc.) to the operator **7000**.

[0052] The control system **300** may include software modules stored in memory for local control, maintenance, installation, commissioning, and for integration into surface control for remote operation. In addition, the control system **300** may include software modules for locating a drill site within a raise **5100** or stope **5000**, locking drilling equipment **8000** in place, powering the drill bit of the drilling equipment **8000**, and communicating the progress of drilling operations to the operator **7000** via the communications network **351** and user interface **310**.

[0053] Advantageously, the control system **300** allows for controlling the modular platform system **100**, its components (e. g., **260**, **580**, etc.), and mobile (or stationary) equipment **8000** by and operator **7000** from a location distant from the raise **5100** and stope **5000** and even from a location distant from the drift **6100**, chamber **6200**, and mine **6000** itself such as at the surface **6400**. As

such, the operator **7000** need not venture near the raise opening **5110** during mining operations.
[0054] According to one embodiment, at least one the first, second, third, and fourth cables **410**, **420**, **430**, **440** (collectively **400**) may include electrical power and communications cables for providing the equipment deck **500** (or **1500**) with electrical power and communications capability and thereby allowing the equipment **8000** to be operated remotely by the control system **300** when the equipment deck **500** (or **1500**) is in the lowered position **502** as described further below. According to one embodiment, the equipment deck **500** (or **1500**) may optionally include a stand-alone power supply **580** such as one or more batteries. According to one embodiment, the wired or wireless communications network **351** of the control system **300** may be extended to the equipment deck **500** (or **1500**) whether in its raised **501** or lowered **502** positions.
[0055] FIG. **7** is a front perspective view illustrating the equipment deck **500** of the modular platform system **100** of FIG. **1** in accordance with an embodiment of the application. FIG. **8** is a top view thereof. And, FIG. **9** is a front view thereof, the rear, left side, and right side views being substantially the same.
[0056] The equipment deck **500** includes: a top deck **510** having an upper surface **513**, a lower surface **514**, and at least one side wall (or side) **511** having an outer surface **515** and an inner surface **516**; a work platform **520** having an upper surface **523** and a lower surface **524**, the lower surface **524** attached (e.g., one or more of attached, connected, joined, bolted, welded, screwed, pinned, hinged, etc.) to the upper surface **513** of the top deck **510**; a bottom deck **530** having an upper surface **533**, a lower surface **534**, and at least one side wall (or side) **531** having an outer surface **335** and an inner surface **536**; first, second, third, and fourth elongate struts **541**, **542**, **543**, **544** spacedly attached between the lower surface **514** of the top deck **510** and the upper surface **533** of the bottom deck **530** to attach and space the top deck **510** to and from the bottom deck **530**; first, second, third, and fourth lift/anchor points **551**, **552**, **553**, **554** spacedly mounted to the outer surface **515** of the side wall **511** of the top deck **510** proximate the first, second, third, and fourth elongate struts **541**, **542**, **543**, **544**, respectively; and, first, second, third, and fourth guide wheels (or guide wheel pairs) **561**, **562**, **563**, **564** mounted to the first, second, third, and fourth struts **541**, **542**, **543**, **544**, respectively. The guide wheels **561**, **562**, **563**, **564** may be rubber guide wheels (or have rubber tires).
[0057] The equipment deck **500** (or **1500**) is sized, shaped, and configured to mate with the top platform **200** via the access opening **250** formed in the top platform **200**. Accordingly, according to one embodiment, the top deck **510** and the bottom deck **520** may be circular in shape as shown in FIGS. **7** to **9**. As shown in FIG. **8**, according to one embodiment, the equipment deck **500** (or **1500**) may measure approximately 1850 mm in radius to fit a raise diameter **5103** of approximately 4000 mm.
[0058] The outer surface **515** of the side wall (or side) **511** of the top deck **510** may have first, second, third, and fourth stabilizers **571**, **572**, **573**, **574** spacedly mounted thereon for improving stability of the equipment deck **500** when docked in the top platform **200** when in the raised position **501**. The stabilizers **571**, **572**, **573**, **574** may provide a closer fit between the outer surface **515** of the top deck **510** and the inner side walls **212**, **222**, **232**, **242** of the front, rear, right, and left platforms **210**, **220**, **230**, **240** of the top platform **200**. As shown in FIG. **8**, the first, second, third, and fourth stabilizers **571**, **572**, **573**, **574** may be spaced 90 degrees (or approximately 90 degrees) apart around the circumference (i.e., side wall **511**) of the top deck **510** (i.e., between the first, second, third, and fourth lift/anchor points **551**, **552**, **553**, **554**.)
[0059] As shown in FIGS. **7** to **9**, the first, second, third, and fourth elongate struts **541**, **542**, **543**, **544** may be sequentially spaced 75 degrees (or approximately 75 degrees), 105 degrees (or approximately 105 degrees), 75 degrees (or approximately 75 degrees), and 105 degrees (or approximately 105 degrees) apart, respectively, around the circumference (i.e., side walls **511**, **531**) of the top and bottom decks **510**, **530**.
[0060] Also as shown in FIGS. **7** to **9**, the first, second, third, and fourth lift/anchor points **551**, **552**,

553, **554** may be sequentially spaced 75 degrees (or approximately 75 degrees), 105 degrees (or approximately 105 degrees), 75 degrees (or approximately 75 degrees), and 105 degrees (or approximately 105 degrees) apart, respectively, around the circumference (i.e., side wall **511**) of the top deck **510** and may be vertically aligned with the first, second, third, and fourth elongate struts **541**, **542**, **543**, **544**, respectively.

[0061] The first, second, third, and fourth guide wheels (or guide wheel pairs) **561**, **562**, **563**, **564** may be pin or axle mounted to respective elongate struts **541**, **542**, **543**, **544** between the top and bottom decks **510**, **520** and are adapted and oriented to engage the side wall **5130** of the raise **5100** when the equipment deck **500** is moved between the raised and lowered positions **501**, **502** as further described below.

[0062] Referring again to FIG. **4**, the first, second, third, and fourth synchronized winches **261**, **262**, **263**, **264** may be sequentially spaced 75 degrees (or approximately 75 degrees), 105 degrees (or approximately 105 degrees), 75 degrees (or approximately 75 degrees), and 105 degrees (or approximately 105 degrees) apart, respectively, around the circumference (i.e., inner walls **212**, **222**, **223**, **242**) of the access opening **250** in the top platform **200** and may be vertically aligned with the first, second, third, and fourth lift/anchor points **551**, **552**, **553**, **554** of the equipment deck **500** (or **1500**), respectively.

[0063] Advantageously, the 105 degree spacing between second and third synchronized winches **262**, **263** and the first and fourth synchronized winches **261**, **264** allows for improved and enlarged access to the equipment deck **500** (or **1500**) (i.e., when in the raised position **501**) for equipment **8000** via the ramps **235**, **245**.

[0064] Referring again to FIGS. **1** to **3**, respective lower ends **411**, **421**, **431**, **441** of the first, second, third, and fourth cables **410**, **420**, **430**, **440** of the first, second, third, and fourth synchronized winches **261**, **262**, **263**, **264** are coupled (e.g., attached by hook, spice, knot, etc.) to the first, second, third, and fourth lift/anchor points **551**, **552**, **553**, **554** of the equipment deck **500** (or **1500**), respectively. The synchronized winches **261**, **262**, **263**, **264** raise **501** the equipment deck **500** (or **1500**) by pulling in (winding up) the respective lower ends **411**, **421**, **431**, **441** of the first, second, third, and fourth cables **410**, **420**, **430**, **440** and lower **502** the equipment deck **500** (or **1500**) by letting out (winding out) the respective lower ends **411**, **421**, **431**, **441** of the first, second, third, and fourth cables **410**, **420**, **430**, **440**.

[0065] According to one embodiment, the equipment deck **500** (or **1500**) may have two top decks **510** (or **1600**) (i.e., the equipment deck **500** (or **1500**) may be a double deck) or the equipment deck **500** (or **1500**) may be a telescoping deck to facilitate two level drilling and blasting.

[0066] According to one embodiment, the top platform **200** and/or the equipment deck **500** (or **1500**) may be equipped with forklift pockets and easy lifting lugs for improved mobility. According to one embodiment, the top platform **200** and/or the equipment deck **500** (or **1500**) may make use of modular components such as power units or power supplies **580** with plug and play cabling. Electrical components may be IP 67 rated. And, services for components (e.g., **260**) and equipment **8000**, including hydraulics, pneumatics, and electrical, may be housed in separate panels or enclosures.

[0067] FIG. **10** is a front perspective view illustrating the modular platform system **100** of FIG. **1** with the equipment deck **500** (or **1500**) thereof shown in a raised position **501** in accordance with an embodiment of the application. And, FIG. **11** is a front perspective view illustrating the modular platform system **100** of FIG. **1** with the equipment deck **500** (or **1500**) thereof shown in a lowered position **502** in accordance with an embodiment of the application.

[0068] Referring to FIGS. **10** and **11**, the synchronized winches **261**, **262**, **263**, **264** (collectively **260**) are operable to move the equipment deck **500** (or **1500**) between a raised position **501** where the upper surface **523** of the work platform **520** of the equipment deck **500** is level (or approximately level) with the upper surface **205** of the top platform **200** proximate the opening **5120** of the raise **5100** (i.e., at the top **5101** of the raise **5100**) in the floor **6300** of the drift **6100** or chamber **6200** of the mine **6000** and a lowered position **502** where the upper surface **523** of the work platform **520** of the equipment deck **500** is positioned proximate the bottom **5102** of the raise **5100** above the top of or within the stope **5000**.

[0069] Advantageously, when in the raised position **501**, the equipment deck **500** (or **1500**) functions to close the access opening **250** for the safety of operators **7000**.

[0070] In operation, the top platform **200** of the modular platform system **100** is assembled over a raise opening **5120** of a raise **5100** in a drift **6100** or chamber **6200** of an underground mine **6000**. The lower ends **411**, **421**, **431**, **441** of the first, second, third, and fourth cables **410**, **420**, **430**, **440** of the first, second, third, and fourth synchronized winches **261**, **262**, **263**, **264** are coupled to the first, second, third, and fourth lift/anchor points **551**, **552**, **553**, **554** of the equipment deck **500** and the equipment deck **500** is docked in the raised position **501** in the access opening **250** of the top platform **200** at the top **5101** of the raise **5100**. First equipment, for example, a mobile drill (or mobile drilling equipment) **8000** is moved via local or remote control by the control system **300** from the surface **6300** of the drift **6100** or chamber **6200** up one of ramps **235**, **245** of the top platform **200** and onto the work platform **520** of the equipment deck **500**. The equipment deck **500** is then lowered down the raise **5100** by the synchronized winches **461**, **462**, **463**, **464** under the control of the control system **300** to a lowered position **502** at the bottom **5102** of the raise **5100** above or at the stope **5000**. While in the lowered position **502**, the mobile drill **8000** is controlled by the control system **300** to perform drilling operations, the progress of which may be viewed by CCTV cameras by an operator **7000** positioned locally at the top platform **200** or remotely from the drift **6100**, chamber **6200**, or even the underground mine **6000** (e.g., from the surface **6400** of the mine **6000**). When drilling operations are completed, the equipment deck **500** is raised up the raise **5100** by the synchronized winches **461**, **462**, **463**, **464** under control of the control system **300** and is returned to its raised position **501** docked in the access opening **250** of the top platform **200**. The mobile drill **8000** may then be moved off of the work platform **520** of the equipment deck **500**, down one of the ramps **235**, **245** of the top platform **200**, and back onto the surface **6300** of the drift **6100** or chamber **6200**. Similar operations may then be followed for second equipment, for example, a mobile emulsion loader (or mobile blast equipment) **8000**. While in the lowered position **502**, the mobile emulsion loader **8000** is controlled by the control system **300** to perform explosives loading operations, that is, to load the holes bored by the drilling operations in the side wall **5130** of raise **5100** or stope **5000** with explosives (or emulsion) in preparation for blasting operations. When explosives loading operations are completed, the mobile emulsion loader **8000** is returned to the drift **6100** or chamber **6200** using the modular platform system **100** as described above and blasting operations may be remotely initiated using the control system **300**. This process may then be repeated for the next ring of the stope **5000**. When ore in the stope **5000** is depleted, the modular platform system **100** may be disassembled and moved along the drift **6100** to the location of the next raise **5100** and stope **5000** where the modular platform system **100** may be reassembled and the entire process may be repeated.

[0071] FIG. **12** is a front perspective view illustrating an equipment deck **1500** for the modular platform system **100** of FIG. **1** having a removable cassette deck **1700** and with the removable cassette deck **1700** thereof shown in an inserted position **1707** in accordance with an embodiment of the application. FIG. **13** is a bottom perspective view thereof. FIG. **14** is a front perspective view illustrating the equipment deck **1500** of FIG. **12** with the removable cassette deck **1700** thereof shown in a removed position **1708** in accordance with an embodiment of the application. FIG. **15** is a bottom perspective view thereof. FIG. **16** is a front view thereof. FIG. **17** is a rear view thereof. FIG. **18** is a top view thereof. FIG. **19** is a bottom view thereof. FIG. **20** is a right side view thereof, the left side view being substantially a mirror image thereof. FIG. **21** is a front perspective view illustrating the top deck **1600** of the equipment deck **1500** of FIG. **12** with the removable cassette deck **1700** thereof removed **1708** and with the top deck **1600** shown in an unfolded position **1607** in accordance with an embodiment of the application. FIG. **22** is a front perspective view illustrating the top deck **1600** of the equipment deck **1500** of FIG. **21** with the removable cassette deck **1700** thereof removed **1708** and with the top deck **1600** shown in a folded position **1608** in accordance with an embodiment of the application. And, FIG. **23** is a right side view thereof, the left side view being substantially a mirror image thereof.

[0072] According to another embodiment of the application, the

[0073] equipment deck **1500** includes a top deck **1600** and a removeable cassette deck or cassette **1700**. Mobile and/or stationary equipment **8000** may be mounted on the cassette **1700** which may then be inserted into the top deck **1600** and the equipment deck **1500** may then be operated in a manner similar to the equipment deck **500** of FIG. **7** with any necessary height adjustments made to the synchronized winches **261**, **262**, **263**, **264** of FIG. **3**. The equipment deck **1500** may include additional equipment decks **1500**, top decks **1600**, or cassettes **1700** spacedly mounted below or above the equipment deck **1500**, top deck **1600**, or cassette **1700**, respectively, depending on the nature of the equipment **8000** to be accommodated.

[0074] According to one embodiment, the equipment deck **1500** includes: a top deck **1600** having an upper surface **1605**, a lower surface **1606**, and at least one side wall (or side) **1611** **1621**, **1631** having an outer surface **1615**, **1625**, **1635**, the top deck **1600** having a slot (or open-sided slot, channel, recess, opening, indent, notch, etc.) **1640** formed therein for receiving a removable cassette deck or cassette **1700**, the top deck **1600** having at least one side wall (or side) (or portion thereof) **1612**, **1622**, **1632** proximate to and defining the slot **1640**; first, second, third, and fourth lift/anchor points **1551**, **1552**, **1553**, **1554** (similar to **551**, **552**, **553**, **554**) spacedly mounted to the outer surfaces **1615**, **1625**, **1635** of the side walls **1611**, **1621**, **1631** of the top deck **1600**; and, first, second, third, and fourth guide wheels (or guide wheel pairs) **1561**, **1562**, **1563**, **1564** (similar to **561**, **562**, **563**, **564**) mounted to the upper and/or lower surfaces **1605**, **1606** of the top deck **1600** proximate to the first, second, third, and fourth lift/anchor points **1551**, **1552**, **1553**, **1554**, respectively. The guide wheels **1561**, **1562**, **1563**, **1564** may be rubber guide wheels (or have rubber tires).

[0075] According to one embodiment, the top deck **1600** has a front end **1601**, a rear end **1602**, a right side **1603**, a left side **1604**, an upper or top surface **1605**, and a lower or bottom surface **1606**. The top platform **1600** includes a front or middle deck **1610**, a right or first side deck **1620**, and a left or second side deck **1630**. The front deck **1610** has an outer side wall (or side) **1611**, an inner side wall (or side) **1612**, a first or right side wall (or side) **1613**, and a second or left side wall (or side) **1614**. The right side deck **1620** has an outer side wall (or side) **1621** and an inner side wall (or side) **1622**. Similarly, the left side deck **1630** has an outer side wall (or side) **1631** and an inner side wall (or side) **1632**. Note that when installed in the modular platform system **100**, the right side deck **1620** is typically positioned adjacent to the front deck **210**.

[0076] The right side **1613** of the front deck **1610** may be removably attached (e.g., coupled, bolted, fastened, pinned, etc.) to the inner side **1622** of the right side deck **1620** proximate a front end **1623** thereof and the left side **1614** of the front deck **1610** is removably attached to the inner side **1632** of the left side deck **1630** proximate a front end **1633** thereof. The inner sides **1622**, **1632** of the side decks **1620**, **1630** are longer than the left and right sides **1613**, **1614** of the front deck **1610** and, as such, when the front, right side, and left side decks **1610**, **1620**, **1630** are attached, the slot **1640** is formed in the top deck **1600** (i.e., defined by the respective inner sides **1612**, **1622**, **1632** of the front, right side, and left side decks **1610**, **1620**, **1630**). According to one embodiment, the slot **1640** may be elongate and generally rectangular in shape.

[0077] According to one embodiment, the right and left side decks **1620**, **1630** may be rotatably attached to the front deck **1610** by respective hinges **1651**, **1652** (or the like) allowing the top deck **1600** to be folded as shown in FIGS. **21** to **23**. In particular, the right and left side decks **1620**, **1630** are movable between respective lowered positions **1628**, **1638** and respective raised positions **1629**, **1639** hence allowing the top deck **1600** to move between respective unfolded and folded positions **1607**, **1608**. Advantageously, in the folded position **1608**, the top deck **1600** and the equipment deck **1500** may be more easily transported.

[0078] According to another embodiment, the front, right side, and left side decks **1610**, **1620**, **1630** may be rigidly attached. According to a further embodiment, the top deck **1600** may be formed in one piece and/or as one unit.

[0079] According to one embodiment, the cassette deck or cassette **1700** has a front end **1701**, a rear end **1702**, a right side **1703**, a left side **1704**, an upper or top surface **1705**, and a lower or bottom surface **1706**. The cassette **1700** has an outer side wall (or side) **1711**, an inner side wall (or side) **1712**, a first or right side wall (or side) **1713**, and a second or left side wall (or side) **1714**. According to one embodiment, the cassette **1700** is elongate and generally rectangular in shape and is sized and shaped to engage and mate with the slot **1640** formed in the top deck **1600**. The front end **1701** of the cassette **1700** may be inserted **1707** into the slot **1640** at the rear end **1602** of the top deck **1640** as shown in FIGS. **12** to **13** or removed **1708** from the slot **1640** of the top deck **1640** as shown in FIGS. **14** to **15**.

[0080] According to one embodiment, the inner side walls **1612**, **1622**, **1632** of the front, right side, and left side decks **1610**, **1620**, **1630** of the top deck **1600** proximate the slot **1640** (i.e., proximate the rear ends **1624**, **1634** of the right and left side decks **1620**, **1630**) have a respective tongue (or raised portion) **1617**, **1627**, **1637** formed thereon for engaging and mating with a respective groove (or lowered portion or channel) **1717**, **1727**, **1737** formed on the inner, right side, and left side walls **1712**, **1713**, **1714** of the cassette **1700** to secure the cassette **1700** within the slot **1640** of the top deck **1600** when the cassette **1700** is moved between the removed position **1708** and the inserted position **1707**.

[0081] FIG. **24** is a front perspective view illustrating the equipment deck **1500** of FIG. **14** with the removable cassette deck **1700** thereof shown in a removed position **1708** and with equipment **8000** mounted on the removable cassette deck **1700** in accordance with an embodiment of the application. And, FIG. **25** is a front perspective view illustrating the equipment deck **1500** of FIG. **12** with the removable cassette deck **1700** thereof shown in an inserted position **1707** and with equipment **8000** mounted on the removable cassette **1700** in accordance with an embodiment of the application.

[0082] According to one embodiment, the mobile and/or stationary equipment **8000** may include a first component **8100** and a second component **8200**. For example, the equipment **8000** may be a drilling system **8000** having a power supply component **8100** (e.g., battery, engine, fuel tank, local control system, communication system, etc.) and a drill component **8200**. The power supply component **8100** may be mounted on the top surface **1705** of the cassette deck **1700** while the drill component **8200** may be mounted on the bottom surface **1706** of the cassette deck **1700** (or vise versa). Mobile and/or stationary equipment **8000** comprising a single component (e.g., **8100** or **8200**) may be mounted on either the top or bottom surfaces **1705**, **1706** of the cassette deck **1700** as required.

[0083] In operation, first equipment (e.g., **8000**) is mounted or pre-mounted on a first cassette deck (e.g., **1700**) and second equipment (e.g., **8000**) is mounted or pre-mounted on a second cassette deck (e.g., **1700**). Similar to the operations described above, the equipment deck **1500** is docked or positioned in (or above as required) the raised position **501** in the access opening **250** of the top platform **200** at the top **5101** of the raise **5100**. The first cassette deck **1700** having first equipment, for example, a stationary drill (or stationary drilling equipment) **8000** mounted thereon is inserted **1707** into the slot **1640** of the top deck **1600** of the equipment deck **1500**. The equipment deck **1500** is then lowered down the raise **5100** by the synchronized winches **461**, **462**, **463**, **464** under the control of the control system **300** to a lowered position **502** at the bottom **5102** of the raise **5100** above or at the stope **5000**. While in the lowered position **502**, the stationary drill **8000** is controlled by the control system **300** to perform drilling operations, the progress of which may be viewed by CCTV cameras by an operator **7000** positioned locally at the top platform **200** or remotely from the drift **6100**, chamber **6200**, or even the underground mine **6000** (e.g., from the surface **6400** of the mine **6000**). When drilling operations are completed, the equipment deck **1500** is raised up the raise **5100** by the synchronized winches **461**, **462**, **463**, **464** under control of the control system **300** and is returned to its raised position **501** (or thereabove as required) docked in the access opening **250** of the top platform **200**. The first cassette deck **1700** may then be removed **1708** from the slot **1640** of the top deck **1600** of the equipment deck **1500**. Similar operations may then be followed for second equipment **8000**, for example, a stationary emulsion loader (or stationary blast equipment) **8000**. In particular, the second cassette deck **1700** having second equipment, for example, a stationary emulsion loader (or stationary blast equipment) **8000** mounted thereon is inserted **1707** into the slot **1640** of the top deck **1600** of the equipment deck **1500**. While in the lowered position **502**, the stationary emulsion loader **8000** is controlled by the control system **300** to perform explosives loading operations, that is, to load the holes bored by the drilling operations in the side wall **5130** of raise **5100** or stope **5000** with explosives (or emulsion) in preparation for blasting operations. When explosives loading operations are completed, the second cassette deck **1700** with the stationary emulsion loader **8000** is returned to the drift **6100** or chamber **6200** using the modular platform system **100** as described above and blasting operations may be remotely initiated using the control system **300**. This process may then be repeated for the next ring of the stope **5000**. When ore in the stope **5000** is depleted, the modular platform system **100** may be disassembled and moved along the drift **6100** to the location of the next raise **5100** and stope **5000** where the modular platform system **100** may be reassembled and the entire process may be repeated.

[0084] According to one embodiment, the equipment deck **1500** of FIGS. **12** to **25** may be used as a substitute for or interchangeably with the equipment deck **500** of FIGS. **1** to **11**. That is, the cassette deck **1700** may be inserted into the slot **1640** of the top deck **1600** and the equipment deck **1500** may be used to transport mobile and/or stationary equipment **8000** loaded onto the upper surfaces **1605**, **1705** of the top deck **1600** and cassette deck **1700**. According to other embodiments, the cassette deck **1700** and mating slot **1640** formed in the top deck **1600** may have any suitable shape (e.g., rectangular, square, semi-circular, circular, etc.).

[0085] According to one embodiment, there is provided a modular platform system **100** for vertical ore stope mining, comprising: a top platform **200** having an access opening **250** formed therein for accessing a raise opening **5120** of a raise **5100** above a stope **500** of a mine **6000**; an equipment deck **500**, **1500** adapted to receive equipment (e.g., mobile equipment and/or stationary equipment) **8000**; and, at least one winch **261**, **262**, **263**, **264** (collectively **260**) having at least one respective cable **410**, **420**, **430**, **440** coupled thereto, the at least one winch **261**, **262**, **263**, **264** mounted on the top platform **200**, the at least one cable **410**, **420**, **430**, **440** (collectively **400**) extendable through the access opening **250**, the at least one cable **410**, **420**, **430**, **440** having at least one respective cable end **411**, **421**, **431**, **441** coupled to the equipment deck **500**, **1500**; wherein the at least one winch **261**, **262**, **263**, **264** is operable to move the equipment deck **500**, **1500** and the equipment (mobile and/or stationary) **8000** between raised position **501** proximate a top **5101** of the raise **5100** at the top platform **200** and a lowered position **502** proximate a bottom **5102** of the raise **5100** at a top of or within the stope **5000**.

[0086] In the above modular platform system **100**, the mine may be an underground mine **6000**. The top platform **200** may be adapted to be moved between raises **5000** in a drift **6100** or chamber **6200** of the underground mine **6000**. The top platform **200** may include a front deck **210**, a rear deck **220**, a first side deck **230**, and a second side deck **240** and the front, rear, first side, and second side decks **210**, **220**, **230**, **240** may be removably attached together to thereby allow the top platform **200** of the modular platform system **100** to be disassembled, relocated, and reassembled while in the drift **6100** or chamber **6200** of the underground mine **6000**. The modular platform system **100** may further include a control system **300** for controlling components (e.g., **261**, **262**, **263**, **264**) of the modular platform system **100**, the at least one winch **261**, **262**, **263**, **264**, and the equipment **8000**. The control system **100** may support local and remote control of the components (e.g., **260**, **580**, etc.) of the modular platform system **100**, the at least one winch **261**, **262**, **263**, **264**, and the equipment **8000**. The control system **300** may have a local user interface or control panel mounted **310** on the top platform **200** for local control. The control system **300** may have a remote user interface or control panel **310** located at a remote location at a distance (e.g., at the surface **6400** of the mine **6000**) from the raise **5100**, drift **6100**, chamber **6200**, or mine **6000** for remote control. The modular platform system **100** may further include a wired or wireless network **351** communicatively coupling the control system **300** to the components (e.g., **260**, **580**, etc.) of the modular platform system **100**, the at least one winch **261**, **262**, **263**, **264**, and the equipment **8000**. The equipment **8000** may include at least one of mobile or stationary drilling equipment and mobile or stationary blasting equipment. The equipment deck **500**, **1500** may include a power supply **580** for providing power to the equipment **8000**. The power supply **580** may be electrically coupled to the top platform **200** via power cabling. The one or more winches **260** may be four synchronized winches **261**, **262**, **263**, **264**. The equipment deck **500** may include a top deck **510** mounted over and spaced from a bottom deck **530** by one or more elongate struts **541**, **542**, **543**, **544**, a work platform **520** mounted over the top deck **510** for receiving the equipment **8000**, and one or more respective guide wheels **561**, **562**, **563**, **564** mounted to the one or more elongate struts **541**, **542**, **543**, **544** between the top deck **510** and the bottom deck **530** and adapted and oriented to engage a side wall **5130** of the raise **5100**. The access opening **250** and equipment deck **500**, **1500** may be circular or approximately circular in shape and the equipment deck **500**, **1500** may be sized to dock and mate with the access opening **250** when in the raised position **501**. The modular platform system **100** may further include one or more ramps **235**, **245** removably attached to the top platform **200** for providing access to the equipment deck **500**, **1500** by mobile equipment **8000**. The modular platform system **100** may further include one or more stabilizers **571**, **572**, **573**, **574** spacedly mounted on an outer surface (e.g., **515**, **1615**, **1711**) of the top deck **510**, **1600** of the equipment deck **500**, **1500** for stabilizing the equipment deck **500**, **1500** in the access opening **250** of the top platform **200** when in the raised position **501**. The power supply **580** may be an optional battery. The equipment deck **1500** may include a top deck **1600** having a slot **1640** formed therein for receiving a removable cassette deck **1700**, the removable cassette deck **1700** for receiving the equipment **8000**, and one or more respective guide wheels **1561**, **1562**, **1563**, **1564** mounted to the top deck **1600** and adapted and oriented to engage a side wall **5130** of the raise **5100**. And, the equipment **8000** may include mobile equipment and/or stationary equipment.

[0087] The above embodiments may contribute to an improved modular platform system **100** for vertical ore stope mining and may provide one or more advantages. First, use of the modular platform system **100** improves operator **7000** safety, increases production and efficiency of mining operations, improves drill and blast accuracy, and provides a skeleton or framework on which other or future automated underground mining equipment may be based. Second, use of the modular platform system **100** decreases ventilation requirements for underground mines **6000**. Third, use of the modular platform system **100** increases safety for workers, miners, and operators **7000** via decreased exposure to geotechnically unstable environments, decreased respiratory particulate matter exposure in the mine **6000**, decreased exposure to blasting, and remote operation of the modular platform system **100** and mobile and/or stationary equipment **8000** from the surface **6400**. Fourth, use of the modular platform system **100** provides increased efficiency and productivity due to automation of mobile and/or stationary equipment **8000**. Fifth, use of the modular platform system **100** provides increased accuracy of drilling and blasting due to automation of mobile and/or stationary equipment **8000**. Sixth, use of the modular platform system **100** provides for reduced size of required excavation. Seventh, use of the modular platform system **100** allows for decreased infrastructure setup. Eighth, the modular design of the modular platform system **100** eases equipment transport and allows the modular platform system **100** to be easily relocated from stope **5000** to stope **5000**. Ninth, the modular platform system **100** may be quickly assembled. Tenth, the modular platform system **100** provides built-in safety features for operators **7000**. And, eleventh,

the modular platform system **100** may handle both mobile and stationary equipment **8000**.
[0088] The embodiments of the application described above are intended to be exemplary only. Those skilled in this art will understand that various modifications of detail may be made to these embodiments, all of which come within the scope of the application.

Claims

1. A modular platform system for vertical ore stope mining, comprising: a top platform having an access opening formed therein for accessing a raise opening of a raise above a stope of a mine; an equipment deck adapted to receive equipment; and, at least one winch having at least one respective cable coupled thereto, the at least one winch mounted on the top platform, the at least one cable extendable through the access opening, the at least one cable having at least one respective cable end coupled to the equipment deck; wherein the at least one winch is operable to move the equipment deck and the equipment between a raised position proximate a top of the raise at the top platform and a lowered position proximate a bottom of the raise at a top of or within the stope.
2. The modular platform system of claim 1, wherein the mine is an underground mine.
3. The modular platform system of claim 2, wherein the top platform is adapted to be moved between raises in a drift or chamber of the underground mine.
4. The modular platform system of claim 3, wherein the top platform includes a front deck, a rear deck, a first side deck, and a second side deck and wherein the front, rear, first side, and second side decks are removably attached together to thereby allow the top platform of the modular platform system to be disassembled, relocated, and reassembled while in the drift or chamber of the underground mine.
5. The modular platform system of claim 1, further comprising a control system for controlling components of the modular platform system, the at least one winch, and the equipment.
6. The modular platform system of claim 5, wherein the control system supports local and remote control of the components of the modular platform system, the at least one winch, and the equipment.
7. The modular platform system of claim 5, wherein the control system has a local user interface or control panel mounted on the top platform for local control.
8. The modular platform system of claim 5, wherein the control system has a remote user interface or control panel located at a remote location at a distance from the raise, drift, chamber, or mine for remote control.
9. The modular platform system of claim 5, further comprising a wired or wireless network communicatively coupling the control system to the components of the modular platform system, the at least one winch, and the equipment.
10. The modular platform system of claim 1, wherein the equipment includes at least one of mobile or stationary drilling equipment and mobile or stationary blasting equipment.
11. The modular platform system of claim 1, wherein the equipment deck includes a power supply for providing power to the equipment.
12. The modular platform system of claim 11, wherein the power supply is electrically coupled to the top platform via power cabling.
13. The modular platform system of claim 1, wherein the one or more winches are four synchronized winches.
14. The modular platform system of claim 1, wherein the equipment deck includes a top deck mounted over and spaced from a bottom deck by one or more elongate struts, a work platform mounted over the top deck for receiving the equipment, and one or more respective guide wheels mounted to the one or more elongate struts between the top deck and the bottom deck and adapted and oriented to engage a side wall of the raise.

15. The modular platform system of claim 1, wherein the access opening and equipment deck are circular or approximately circular in shape and wherein the equipment deck is sized to dock and mate with the access opening when in the raised position.
16. The modular platform system of claim 1, further comprising one or more ramps removably attached to the top platform for providing access to the equipment deck by mobile equipment.
17. The modular platform system of claim 14, further comprising one or more stabilizers spacedly mounted on an outer surface of the top deck of the equipment deck for stabilizing the equipment deck in the access opening of the top platform when in the raised position.
18. The modular platform system of claim 11, wherein the power supply is an optional battery.
19. The modular platform system of claim 1, wherein the equipment deck includes a top deck having a slot formed therein for receiving a removable cassette deck, the removable cassette deck for receiving the equipment, and one or more respective guide wheels mounted to the top deck and adapted and oriented to engage a side wall of the raise.
20. The modular platform system of claim 1, wherein the equipment includes mobile equipment and stationary equipment.